

Causality-Preserving Domain Generalization via Adaptive Fourier Mixup for RUL Prediction

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Abstract—Domain generalization (DG) in time series poses significant challenges due to domain shift, particularly under the strict DG setting, where no target domain data are available during training. To address this, we propose AFM-CIR, a unified framework that integrates semantic-similarity-guided Adaptive Fourier Mixing (AFM) with Causality-Inspired Regression (CIR). Specifically, we construct a domain-invariant order-preserving guidance embedding that drives a similarity-based adaptive modulation of amplitude mixing and a bounded shortest-angle phase perturbation, thereby generating label-consistent and causally coherent augmented samples. CIR then enforces invariance and inter-dimensional independence through correlation factorization, while causal sufficiency is encouraged via adversarial masking. We further provide theoretical guarantees of the controllability of phase interventions, supported by mutual information analysis and Lipschitz-spectral norm bounds. Extensive experiments on four widely used benchmark industrial data sets demonstrate that AFM-CIR consistently achieves state-of-the-art performance, outperforming strong ERM, general DG, and task-specific DG baselines

Index Terms—Domain generalization, frequency-domain augmentation, domain shift, causality-inspired learning, remaining useful life prediction.

I. INTRODUCTION

TIME-SERIES, characterized by sequential structure and temporal dependencies, are ubiquitous in many real-world scenarios [1], [2], including finance [3], healthcare [4], and industrial settings [5]. Although deep learning has recently advanced time-series modeling by leveraging large labeled datasets to capture complex temporal dynamics [6], performance often

degrades sharply under domain shift [7], [8], where the data generation process varies across sources, locations, sensing hardware, operating regimes, or temporal conditions. Consequently, models trained on source domains frequently fail to generalize to unseen targets, motivating DG methods that seek representations and predictors robust to distributional variation using only multiple source domains, without access to target-domain data during training [9], [10].

Within the landscape of DG approaches [11], data augmentation stands out as a practical and widely applicable route [12], [13]. By inducing variability in existing samples, it exposes models to a broader range of distributions during training, thereby mitigating overfitting to source domains and enhancing robustness to unseen targets. The Mixup strategy [14], [15] interpolates pairs of sequences in the input or feature space using randomly sampled weights; while conceptually simple, it has consistently demonstrated strong performance across widely used benchmarks. Extending beyond interpolation, generative augmentation [16], [17] adopts learned generative models to synthesize additional sequences from source domains, offering greater flexibility in capturing data variations with the expense of higher computational cost and model complexity.

Although Mixup and generative augmentation expand training support, they often overlook the semantic structures that govern time-series data [18]. Linear interpolation can blur temporal dependencies and distort task semantics, while generative models can synthesize sequences that capture spurious correlations or domain-specific artifacts rather than invariant patterns. In both cases, the failure to disentangle task-relevant information from nuisance variations risks altering the input-label relationship and undermining the stable, domain-invariant structures essential for generalization. These limitations highlight the need for augmentation strategies that preserve semantic consistency while filtering out domain-specific noise.

Building on this line of efforts, causal learning has emerged as a principled and increasingly influential approach to addressing the same challenge. By extracting the semantic causal structure while suppressing spurious non-causal variation, it provides a theoretically grounded path to out-of-distribution (OOD) generalization. Invariant risk minimization (IRM) and subsequent variants promote cross-environment agreement by enforcing gradient invariance and controlling risk variance [19], [20]. In addition, auxiliary function formulations instantiate the principle of independence of causal mechanisms (ICM) to stabilize conditionals across domains and to separate a task-relevant signal from nuisance factors [21]. Meanwhile, graphical

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criterion approaches employ causal graphs with constraint or score-based procedures to encode independent mechanisms and invariance, with invariant causal prediction (ICP) as a representative method [22]. Recently, structural causal models (SCMs) have been introduced to formalize DG with encouraging outcomes, among which causality inspired representation learning (CIRL) [23] represents a notable attempt rooted in causal learning. However, existing CIRL formulations are primarily developed for classification-style settings and do not directly address phase-sensitive time-series regression under DG.

CIRL encourages models to capture stable causal relationships by generating synthetic samples across domains and has demonstrated promising results in image classification. However, in time-series contexts, restricting interventions to amplitude while keeping phase fixed is ill-posed: the phase component governs temporal shifts and alignments that encode the signal's temporal and semantic structure. It is therefore indispensable for learning sequential dynamics [24], [25]. In addition, both the sampling of mixed examples and the interpolation coefficients in CIRL are chosen randomly. Such randomness may inadvertently break label consistency, thereby undermining the stability of the learned representations. These limitations highlight the necessity of developing a causality-preserving and controllable augmentation strategy tailored to time-series settings. Motivated by this gap, we make a task-driven extension beyond CIRL by introducing (i) similarity-guided amplitude mixing with bounded phase intervention for semantic preservation, (ii) a regression-oriented causal representation objective, and (iii) a controllability analysis for the proposed phase perturbation.

Accordingly, we propose the **Adaptive Fourier Mixup with Causal Inspired Regression (AFM-CIR)**, a unified framework for time-series DG that integrates controllable data augmentation with causality-preserving representation learning. The highlights of this work are summarized as follows.

- We design a domain-invariant, order-preserving guidance embedding space that provides a reliable, semantics-aware similarity measure to steer augmentation.
- AFM module combines similarity-guided amplitude mixing with bounded, semantic-preserving phase perturbation, supported by theoretical guarantees from mutual information and Lipschitz–spectral bounds.
- An extension of causality-inspired representation learning to regression, with a factorization objective and an adversarial masking mechanism that together enforce invariance, independence, and causal sufficiency.
- Information-theoretic and mathematical analyzes, along with extensive experiments on publicly available industrial datasets for remaining useful life (RUL) prediction under distribution shift, confirm the effectiveness and robustness of the proposed method.

The remainder of this paper is organized as follows. Section II introduces the relevant preliminaries. Section III presents the proposed AFM-CIR framework in detail. Section IV discusses the necessity of phase perturbation and provides a causal analysis with theoretical guarantees. Section V reports the experimental results and ablation studies. Section VI concludes the paper.

II. PRELIMINARY

A. Notations

1) *Domain Generalization*: Let $\mathcal{S} = \{\mathcal{S}^k\}_{k=1}^K$ denote the K known source domains, where the k -th source domain is $\mathcal{S}^k = \{(\mathbf{x}_i^k, y_i^k)\}_{i=1}^{n_s^k}$, containing n_s^k labeled samples with $\mathbf{x}_i^k \in \mathcal{X}$ as input sequence and $y_i^k \in \mathcal{Y}$ as ground-truth label. The target domain is denoted by $\mathcal{T} = \{(\mathbf{x}_j, y_j)\}_{j=1}^{n_t}$ with n_t samples.

Unlike Unsupervised Domain Adaptation (UDA), which assumes access to unlabeled target data, DG learns a predictive model solely from multiple source domains without prior knowledge of the target distribution. The goal is to learn a predictive function that achieves robust performance on unseen domains:

$$f^* = \arg \min_{f \in \mathcal{H}} \mathbb{E}_{(\mathbf{x}, y) \in \mathcal{T}} [\mathcal{L}(f(\mathbf{x}), y)],$$

where $\mathcal{H}$ denotes the hypothesis space. A typical DG-based time-series prediction model consists of a feature extractor $\hat{g} : \mathcal{X} \rightarrow \mathcal{Z}$ that maps raw sequences to a latent space $\mathcal{Z}$, and a prediction head $\hat{h} : \mathcal{Z} \rightarrow \mathcal{Y}$ producing the final output.

2) *Fourier Transform*: Given an input signal $\mathbf{x}$, its Fourier transform [26] can be expressed in polar form as

$$\mathcal{F}(\mathbf{x}) = \mathcal{A}(\mathbf{x}) e^{j\mathcal{P}(\mathbf{x})},$$

where j is the imaginary unit, $\mathcal{A}(\mathbf{x})$ and $\mathcal{P}(\mathbf{x})$ denote the amplitude and phase components, respectively. Let $\mathcal{R}(\mathbf{x})$ and $\mathcal{I}(\mathbf{x})$ be the real and imaginary parts of $\mathcal{F}(\mathbf{x})$. Then

$$\mathcal{A}(\mathbf{x}) = \sqrt{\mathcal{R}(\mathbf{x})^2 + \mathcal{I}(\mathbf{x})^2},$$

$$\mathcal{P}(\mathbf{x}) = \text{atan2}(\mathcal{I}(\mathbf{x}), \mathcal{R}(\mathbf{x})).$$

Here, atan2 denotes the two-argument arctangent function, which returns the phase angle in radians with the convention $\mathcal{P}(\mathbf{x}) \in (-\pi, \pi]$ throughout this paper. Both the Fourier transform $\mathcal{F}(\cdot)$ and its inverse $\mathcal{F}^{-1}(\cdot)$ can be efficiently computed using the FFT algorithm [27], [28].

B. Mixup-Based Data Augmentation

1) *Time-Domain Mixup*: A straightforward augmentation baseline is to apply linear mixup directly in the time domain [12], [15], [29]. Given two signals $\mathbf{x}$ and $\tilde{\mathbf{x}}$, the interpolated signal is defined as,

$$\mathbf{x}^+ = \lambda \mathbf{x} + (1 - \lambda) \tilde{\mathbf{x}}, \quad \lambda \sim U(0, 1).$$

While effective in vision and speech domains, this method indiscriminately mixes both low-level statistics and high-level semantics. As a result, it often produces unrealistic samples and may produce label-inconsistent samples, which limits its reliability in sensitive applications such as time-series modeling.

2) *Frequency-Domain Mixup*: To overcome the limitations of time-domain interpolation, subsequent work has turned to the Fourier spectrum. Many early studies [30], [31] have shown that the phase component largely encodes high-level semantics, while the amplitude mainly reflects low-level statistics. Building on this observation, FACT [32] perturbs only the amplitude spectrum via linear interpolation between the target signal $\mathbf{x}$

and a randomly sampled source signal $\tilde{\mathbf{x}}$:

$$\mathcal{A}(\mathbf{x}^+) = \lambda \mathcal{A}(\mathbf{x}) + (1 - \lambda) \mathcal{A}(\tilde{\mathbf{x}}),$$

with $\lambda \sim U(\rho, 1)$ controlling the strength of the augmentation by tuning the parameter ρ . The mixed amplitude spectrum is then combined with the original phase to yield a new Fourier representation:

$$\mathcal{F}(\mathbf{x}^+) = \mathcal{A}(\mathbf{x}^+) e^{j\mathcal{P}(\mathbf{x})},$$

which is finally transformed back into the time domain:

$$\mathbf{x}^+ = \mathcal{F}^{-1}(\mathcal{F}(\mathbf{x}^+)).$$

This approach avoids mixing semantic content across domains and has proven effective for image data. However, since it keeps the phase component fixed, it may suffer from limited robustness to phase shifts, which are especially prevalent and critical in time-series signals.

C. Causal Foundations

Causality provides a principled view of how data are generated and why models generalize across domains [33], [34], [35]. Formally, we introduce an SCM to formalize DG; specifically, we assume that each input X is constructed from causal factors S and domain-related non-causal factors U , and that only S causally determines the label Y [23]. From this perspective, two principles are central. The *common cause principle* [36] suggests that correlations often arise from hidden factors; in DG, spurious dependence between X and Y may be driven by domain-specific noise U rather than the true cause S . The *independence of causal mechanisms* (ICM) [37], [38] further states that the causal relation $P(Y | S)$ is stable across domains, while U may vary. Hence, the goal is to capture S and its invariant link to Y , while filtering out U .

A common method in causal learning is to use interventions as a way to generate new data while preserving causal relations [39], [40], [41]. By intervening on the non-causal factor U and forcing it to take a new value u' , the distribution of X is altered but the mapping from the causal factor to the label remains unchanged:

$$P(X, Y | \text{do}(U = u')) = \int P(S) P(X | S, u') P(Y | S) dS,$$

so that $P(Y | S)$ remains invariant under such interventions, where $\text{do}(\cdot)$ denotes the do operator. This general principle justifies spectrum-based perturbations (e.g., Fourier mixing) as interventions on U that enrich $P(X)$ for training while preserving the task-defining relation between S and Y .

Scope of domain shifts: We clarify the type of domain shifts considered in this work. Following the SCM formulation above, we assume that each observation X is generated from a causal degradation factor S and domain-related nuisance factors U , where only S causally determines the target Y . Across domains, we assume an invariant causal mechanism $P(Y | S)$, while U may vary due to operating regimes (e.g., speed/load), sensing conditions, and temporal misalignment/phase drift. Under this assumption, the domain shift primarily appears as variations

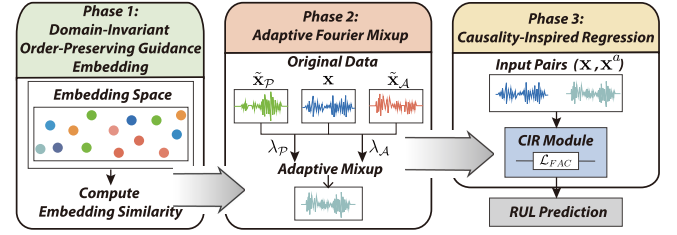


Fig. 1. Overview of the AFM-CIR pipeline. Phase 1 learns a domain-invariant, order-preserving guidance embedding and outputs similarity scores; Phase 2 uses these similarities to set adaptive coefficients for amplitude mixing and bounded phase perturbation to generate augmented samples; Phase 3 trains the predictor with paired original/augmented data under the causality-inspired regression objective.

in the observation distribution induced by U (e.g., changes in spectral amplitude statistics and phase/timing perturbations), which may also induce spurious changes in $P(Y | X)$ in the raw space. Our AFM module is designed to simulate such U -induced shifts via similarity-guided amplitude mixing and bounded phase perturbation, and our CIR objective further enforces representation invariance and inter-dimensional independence under these interventions.

Limitations: We do not claim guarantees in scenarios where the underlying degradation-to-RUL mechanism changes across domains (i.e., $P(Y | S)$ is not invariant), such as new failure modes or altered labeling protocols.

III. METHODOLOGY

A. Pipeline Overview

The proposed AFM-CIR framework is designed to enhance time-series prediction under domain shift by integrating semantics-aware Fourier augmentation with causality-inspired regression, where “semantics-aware” means that both the pairing strategy and the augmentation strength are guided by similarity in the domain-invariant, order-preserving embedding space so that mixed samples remain comparable in degradation state. The pipeline proceeds in three stages, as shown in Fig. 1. First, a lightweight autoencoder is pretrained to construct a domain-invariant, order-preserving embedding space, in which pairwise similarities faithfully reflect temporal and label proximity across heterogeneous domains. This embedding provides semantics-aware guidance for subsequent augmentation. Next, each anchor sequence is paired with two companions: a reference sample for amplitude mixing and a similar neighbor for phase adjustment. Their learned similarities are mapped to adaptive coefficients that govern two complementary Fourier-domain operations, adaptive amplitude mixing and bounded, causality-preserving phase perturbation, meaning that the perturbation is similarity-guided and bounded so as to preserve label semantics, thereby enriching diversity while preserving semantic consistency. Finally, both original and augmented sequences are fed into a causality-inspired regression module that enforces invariance to augmentation, independence across feature dimensions, and causal sufficiency via adversarial masking. Together,

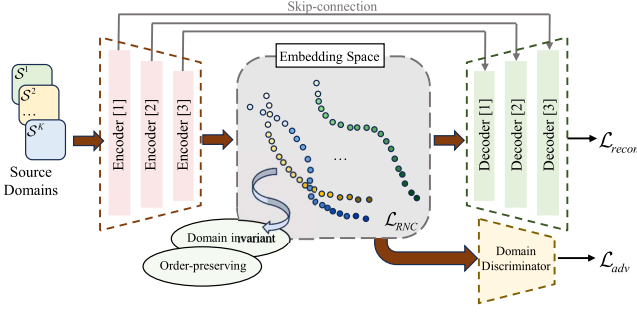


Fig. 2. Overview of domain-invariant order-preserving guidance embedding space.

these components align learned representations with underlying causal mechanisms, thereby improving the robustness and generalization ability of time-series prediction.

B. Phase 1: Domain-Invariant Order-Preserving Guidance Embedding

Conventional mixup strategies, whether in the time domain [12], [29] or the frequency domain [23], [32], [42], typically rely on randomly sampled interpolation weights. Such randomness neglects whether paired signals are semantically comparable, often resulting in unrealistic combinations and distortion of causal structure.

To address this, we construct a *guidance embedding space* that provides a domain-invariant, order-preserving similarity measure across heterogeneous domains, as illustrated in Fig. 2. Pairwise similarities derived from this space are then transformed into interpolation coefficients via a monotonic mapping, which in turn adaptively regulates amplitude mixing and phase perturbation in the Fourier domain.

Concretely, we adopt a lightweight 1D-UNet autoencoder backbone $f(\cdot; \psi)$ that maps a raw vibration signal $\mathbf{x}$ into a latent embedding $\mathbf{z} = f(\mathbf{x}; \psi)$ while reconstructing the input $\hat{\mathbf{x}}$, ensuring that local temporal patterns are preserved. The reconstruction loss is defined as,

$$\mathcal{L}_{recon} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2.$$

Minimizing this reconstruction loss ensures that the embeddings faithfully preserve signal fidelity. To further shape the embedding space, we augment the training objective with two additional terms: a multi-class domain-adversarial loss and an order-preserving contrastive loss. These additions encourage the embedding space to be domain-invariant, i.e., invariant to spurious discrepancies across source domains, while being order-preserving, ensuring that the latent space faithfully reflects the ordinal structure of the target labels.

To enforce domain invariance, we employ a multi-class adversarial loss [43], [44]. A domain discriminator $D(\cdot; \phi)$ predicts the source domain label d_i of $\mathbf{z}_i$, and is trained to minimize the standard cross-entropy loss, while the encoder is trained adversarially via a gradient reversal layer [45]. The loss is

$$\mathcal{L}_{adv} = -\frac{1}{N_s} \sum_{i=1}^{N_s} \sum_{k=1}^K \mathbf{1}[k = d_i] \log D_k(f(\mathbf{x}_i; \psi); \phi),$$

where d_i denotes the true domain label of sample $\mathbf{x}_i$, N_s is the number of all source domain samples, and $D_k(\cdot; \phi)$ is the predicted probability of belonging to the k -th source domain. This adversarial mechanism ensures that the encoder learns a domain-invariant embedding space that generalizes across multiple source domains.

To further incorporate ordinal structure, we employ the Rank-N-Contrast (RNC) loss [46], which leverages continuous labels to construct positive and negative pairs. For anchor i with label y_i , each sample j defines a negative set

$$\mathcal{N}_{i,j} = \{k \mid k \neq i, d_l(y_i, y_k) \geq d_l(y_i, y_j)\},$$

where $d_l(\cdot, \cdot)$ is a label-distance metric. The per-sample RNC loss is

$$\mathcal{L}_{RNC}^i = \frac{1}{B-1} \sum_{\substack{j=1 \\ j \neq i}}^B -\log \frac{\exp(\mathbf{z}_i \cdot \mathbf{z}_j / \tau)}{\sum_{k \in \mathcal{N}_{i,j}} \exp(\mathbf{z}_i \cdot \mathbf{z}_k / \tau)},$$

and the batch-level loss averages over all anchors:

$$\mathcal{L}_{RNC} = \frac{1}{B} \sum_{i=1}^B \mathcal{L}_{RNC}^i,$$

where B is the batch size. This formulation effectively aligns embedding distances with label differences, ensuring that samples with closer labels are embedded nearer together, while those with larger label gaps are separated. This ensures that embedding distances faithfully reflect label distances, producing an order-preserving latent geometry.

The encoder is trained with a weighted combination of reconstruction, adversarial, and RNC losses as follows

$$\mathcal{L}_{pretrain} = \mathcal{L}_{recon} + \alpha_{adv} \mathcal{L}_{adv} + \alpha_{RNC} \mathcal{L}_{RNC}.$$

This yields an embedding space that is simultaneously domain-invariant and order-preserving, furnishing a reliable similarity metric to drive downstream augmentation.

In our implementation, the pretrained guidance encoder is kept *frozen* after Phase 1 and used only to provide a stable, data-dependent similarity metric. We intentionally decouple this metric from downstream regression optimization, since joint fine-tuning would make the embedding space non-stationary, causing neighborhood relations to drift during training and potentially weakening the intended semantics-aware, label-preserving behavior.

C. Phase 2: Adaptive Fourier Mixup

Building on the guidance embedding space, we propose AFM, a Fourier-domain augmentation scheme that enriches sample diversity without disrupting causal structure, as shown in Fig. 3.

Concretely, pairwise similarities computed in the embedding space are transformed into *adaptive mixup rates*, which regulate two complementary operations: *cross-domain amplitude mixing* and *phase perturbation*. Each operation draws from its own sampling pool, and the adaptive rates ensure that augmentation strength varies smoothly with degradation proximity.

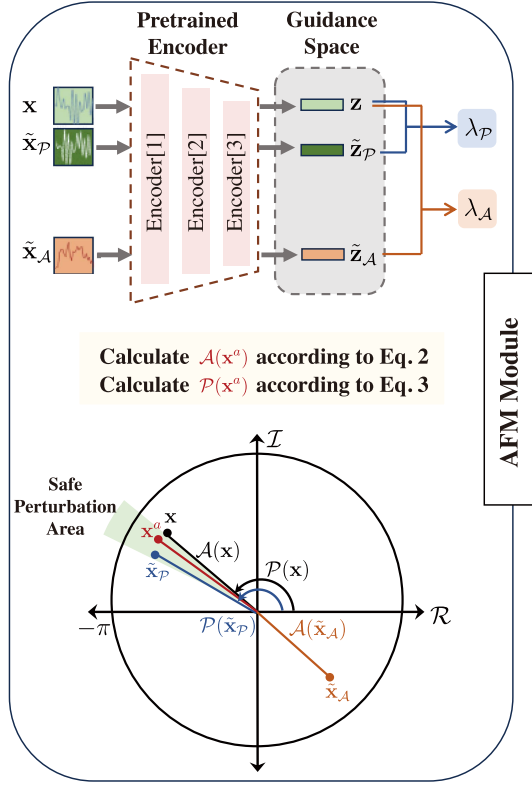


Fig. 3. Illustration of the proposed AFM module. The pretrained encoder maps raw signals into the guidance embedding space, where pairwise similarities are transformed into adaptive coefficients (λ_A, λ_p). These coefficients regulate cross-domain amplitude mixing and causality-preserving phase perturbation in the Fourier domain. The phasor diagram (bottom) depicts the safe perturbation area, where the anchor phase $\mathcal{P}(x)$ is adjusted adaptively toward the phase of neighbor sample $\tilde{x}_p$ along the minimal angular gap, ensuring that perturbations remain physically plausible and semantics-preserving.

1) *Adaptive Mixup-Rate Amplitude Mixing*: Let the anchor sample be x with domain label $d(x) \in \{1, \dots, K\}$. We first pick a *reference sample* $\tilde{x}_A$ from the union of all source domains except the anchor,

$$\tilde{x}_A \in \left(\bigcup_{k=1}^K \mathcal{S}^k \right) \setminus \{x\}.$$

Then we compute the embedding similarity

$$\sigma_A = \sigma(z, \tilde{z}_A) = \sigma(f(x; \psi), f(\tilde{x}_A; \psi)) \in [0, 1],$$

and convert it into a mixing coefficient via a monotonic mapping:

$$\lambda_A = M(\sigma_A, \gamma_A) = 1 - (1 - \sigma_A) \gamma_A \in (\gamma_A, 1). \quad (1)$$

Here λ_A serves as the *adaptive mixup rate* for amplitude mixing. Higher similarity yields a smaller λ_A , thus assigning more weight to the reference amplitude and enabling stronger cross-domain blending; conversely, lower similarity drives λ_A closer to 1, preserving the anchor amplitude and limiting mixing.

Let $\mathcal{F}(x) = \mathcal{A}(x)e^{j\mathcal{P}(x)}$ and $\mathcal{F}(\tilde{x}_A) = \mathcal{A}(\tilde{x}_A)e^{j\mathcal{P}(\tilde{x}_A)}$, then the augmented amplitude is

$$\mathcal{A}(x^a) = \lambda_A \mathcal{A}(x) + (1 - \lambda_A) \mathcal{A}(\tilde{x}_A). \quad (2)$$

When $\tilde{x}_A$ is close to x in the embedding space, $\lambda_A \rightarrow \gamma_A$ allows substantial cross-domain blending; otherwise $\lambda_A \rightarrow 1$ preserves the anchor. This injects domain-level diversity into the low-order spectrum while respecting degradation proximity.

2) *Adaptive Mixup-Rate Phase Perturbation*: While amplitude mixing diversifies magnitudes, it overlooks phase variability. In the context of industrial time-series, phase distortions (drift, jitter, timing shifts) are pervasive due to load fluctuations, speed variations, and progressive degradation. Ignoring them leaves a critical blind spot and degrades robustness under unseen conditions. We therefore introduce an explicit *phase perturbation* that exposes the model to realistic phase shifts during training while preserving causal semantics, which means that the perturbation is intended to leave the causal factor S and thus the label Y unchanged while only introducing realistic phase variability in the observed signal. Consequently, the predictor acquires invariance to common phase deviations and improved robustness to distribution shifts, as theoretically guaranteed in Section IV and corroborated by ablations in Section V.

The phase $\mathcal{P}(\cdot)$ encodes the high-level causal semantics of the signal, i.e., the label-relevant structure that reflects the underlying degradation factor S . Directly mixing phases would disrupt this structure by inducing destructive interference among misaligned harmonics, thereby corrupting temporal information. To mitigate this risk, we design a *causality-preserving phase perturbation* that is *local*, *semantics-aware*, and *conservatively weighted*, meaning that the phase is adjusted only toward a neighbor with a similar degradation state, and the perturbation magnitude is bounded and smoothly down-weighted as the similarity decreases. As illustrated in Fig. 3, this adjustment is implemented on the phasor plane and restricted to a safe perturbation region, following the shortest angular path toward the selected neighbor.

We instantiate the procedure by first selecting a *neighbor sample* $\tilde{x}_p$ sufficiently similar to the anchor in the guidance embedding:

$$\tilde{x}_p \in \left(\bigcup_{k=1}^K \mathcal{S}^k \right) \setminus \{x\}, \quad \text{s.t. } \sigma(z, \tilde{z}_p) \geq \eta,$$

where $\eta \in (0, 1)$ is a similarity threshold. Among all eligible samples satisfying the constraint, we *uniformly sample one neighbor at random* to avoid deterministic nearest-neighbor pairing and to increase augmentation diversity. We then compute the similarity and obtain the perturbation rate by reusing the same mapping in Eq. (1):

$$\sigma_p = \sigma(z, \tilde{z}_p), \quad \lambda_p = M(\sigma_p, \gamma_p) \in (\gamma_p, 1).$$

Here, λ_p acts as the *adaptive mixup rate* for phase perturbation. Higher similarity results in a smaller λ_p (equivalently, larger $(1 - \lambda_p)$), encouraging a stronger yet bounded phase adjustment toward close neighbors; conversely, lower similarity drives $\lambda_p \rightarrow 1$, yielding negligible perturbation and preserving the anchor phase. The hyperparameter γ_p constrains the minimum perturbation strength to remain physically plausible.

We then perturb the anchor phase by the shortest angular difference toward the neighbor:

$$\theta \equiv [\mathcal{P}(x) - \mathcal{P}(\tilde{x}_p)] \pmod{2\pi},$$

$$\Delta\theta = \begin{cases} \theta - 2\pi, & \text{if } \theta > \pi, \\ \theta, & \text{otherwise,} \end{cases}$$

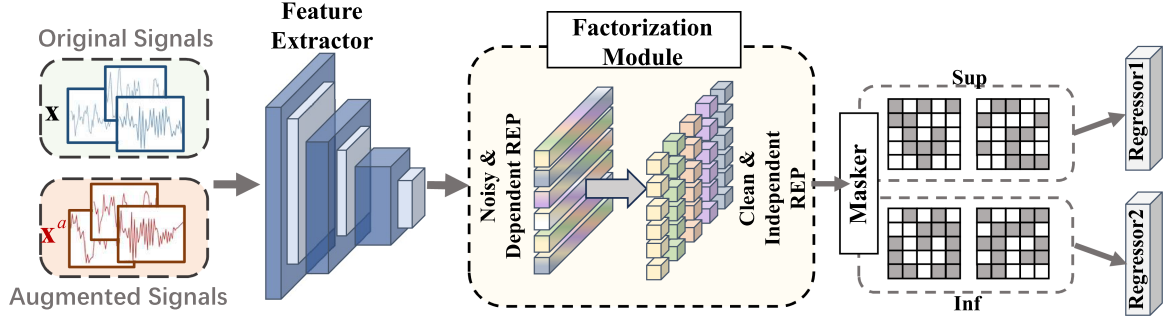


Fig. 4. Overview of the proposed CIR framework. Original and AFM-augmented signals are first encoded into latent representations. The factorization module enforces causal invariance and independence through a correlation-based loss, while the adversarial masker exposes inferior dimensions via a min-max game with the feature extractor, compelling them to encode complementary information and ensuring causal sufficiency for regression.

$$\mathcal{P}(\mathbf{x}^a) = \begin{cases} \mathcal{P}(\mathbf{x}) - |\Delta\Theta|(1 - \lambda_p), & \text{if } \Delta\Theta > 0, \\ \mathcal{P}(\mathbf{x}) + |\Delta\Theta|(1 - \lambda_p), & \text{otherwise.} \end{cases} \quad (3)$$

Here, $\text{sign}(\Delta\Theta)$ specifies the rotation direction: $\Delta\Theta > 0$ corresponds to a counterclockwise adjustment, whereas $\Delta\Theta \leq 0$ corresponds to a clockwise adjustment in the phasor diagram. $|\Delta\Theta|$ is the minimal angular gap in radians, and $1 - \lambda_p \in (0, 1 - \gamma_p)$ provides a bounded, similarity-based adaptive perturbation strength.

3) *Causality-Preserving Augmentation*: The proposed AFM scheme integrates the two complementary operations described above. Adaptive mixup-rate amplitude mixing introduces controlled variability by blending magnitudes across domains, while adaptive mixup-rate phase perturbation regularizes phase shifts in a manner consistent with degradation similarity. Together, they ensure that the causal semantics of temporal progression remain intact.

The augmented signal is finally transformed back to the time domain via inverse Fourier transform,

$$\mathbf{x}^a = \mathcal{F}^{-1}\left(\mathcal{A}(\mathbf{x}^a) e^{j\mathcal{P}(\mathbf{x}^a)}\right),$$

yielding signals that are label-preserving and more diverse for training. For simplicity, the augmented label is kept identical to the anchor, i.e., $y^a = y$, ensuring that augmentation affects only input variability while leaving the regression target unchanged. We refer readers to Sec. IV-A for a focused discussion on why phase perturbation is essential and how it motivates the subsequent causal analysis.

D. Phase 3: Causality-Inspired Regression

Building on the causality-preserving Fourier augmentations $\{(\mathbf{x}, \mathbf{x}^a, y)\}$, we adopt CIRL [23] as the foundation for our regression module. CIRL offers a principled framework for extracting causal-invariant features, but its original design is tailored to classification tasks and relies on amplitude-only augmentation. In contrast, RUL prediction is a regression problem that requires order consistency across continuous labels, and robustness must further account for phase perturbations.

To this end, we propose the CIR by extending CIRL to the regression setting and couple it with our AFM module, thereby

enforcing causal invariance under both original and augmented signals. The whole framework of CIR is shown in Fig. 4.

Concretely, we instantiate a feature extractor $\hat{g}(\cdot)$ that maps time-series into latent representations, two regressor $\hat{h}_1(\cdot)$ and $\hat{h}_2(\cdot)$ dedicated to superior and inferior subsets, and an adversarial masker $\hat{w}(\cdot)$ that operates as a dimension-wise selector, adversarially exposing inferior dimensions to promote causal sufficiency. The training objective follows the *Common Cause Principle* [36] and the *ICM Principle* [37], [38], ensuring that learned representations explain label-relevant variability while remaining independent across dimensions and invariant to non-causal perturbations.

For a mini-batch $\{(\mathbf{x}_i, \mathbf{x}_i^a, y_i)\}_{i=1}^B$, we denote

$$\mathbf{R} = \hat{g}(\mathbf{x}) = [(r_1)^T, \dots, (r_B)^T]^T \in \mathbb{R}^{B \times N},$$

$$\mathbf{R}^a = \hat{g}(\mathbf{x}^a) = [(r_1^a)^T, \dots, (r_B^a)^T]^T \in \mathbb{R}^{B \times N},$$

as the representation matrices of original and augmented samples, respectively. Here, B is the batch size and N is the number of dimensions.

To encourage representations to preserve causal factors that remain stable under AFM, we introduce a correlation-based regularization. Specifically, $COR(\cdot, \cdot)$ is introduced as a generic similarity measure between the i -th dimension of the representation $\mathbf{R}$ and its augmented counterpart $\mathbf{R}^a$. Maximizing this alignment enforces element-wise invariance in the representation space:

$$\max_{\hat{g}} \frac{1}{N} \sum_{i=1}^N COR(\tilde{\mathbf{r}}_i, \tilde{\mathbf{r}}_i^a), \quad (4)$$

where $\tilde{\mathbf{r}}_i$ and $\tilde{\mathbf{r}}_i^a$ denote the Z-score normalized i -th columns of $\mathbf{R}$ and $\mathbf{R}^a$, respectively.

Beyond invariance, the ICM principle further requires that causal factors be mutually independent. To encourage this property, we additionally penalize correlations across distinct dimensions

$$\min_{\hat{g}} \frac{1}{N(N-1)} \sum_{i \neq j} COR(\tilde{\mathbf{r}}_i, \tilde{\mathbf{r}}_j^a). \quad (5)$$

To unify the optimization goal of Eq. (4) and Eq. (5), we build a correlation matrix $\mathbf{C}$ with entries,

$$\mathbf{C}_{ij} = \frac{\langle \tilde{\mathbf{r}}_i, \tilde{\mathbf{r}}_j^a \rangle}{\|\tilde{\mathbf{r}}_i\| \|\tilde{\mathbf{r}}_j^a\|}, \quad i, j \in \{1, \dots, N\},$$

where $\langle \cdot, \cdot \rangle$ denotes the inner product. By construction, diagonal entries $\mathbf{C}_{ii}$ measure per-dimension invariance before and after AFM and are thus encouraged to approach 1, whereas off-diagonal entries $\mathbf{C}_{ij}$ ($i \neq j$) quantify cross-dimension dependence and are encouraged to be close to 0. Based on this intuition, we design a factorized loss $\mathcal{L}_{FAC}$ as follow,

$$\mathcal{L}_{FAC} = \frac{1}{2} \|\mathbf{C} - \mathbf{I}\|_F^2,$$

where $\|\cdot\|_F$ denotes the Frobenius norm. Minimizing $\mathcal{L}_{FAC}$ jointly enforces invariance and independence, transforming noisy, entangled representations into clean and factorized ones, in line with both the Common Causal and ICM principles.

While $\mathcal{L}_{FAC}$ promotes invariance and independence, *causal sufficiency* requires that all dimensions contribute to degradation prediction. To uncover inferior dimensions and compel them to carry more predictive information, we introduce the adversarial masker module.

Specifically, we obtain a k -hot masker $\mathbf{m} \in \{0, 1\}^N$ with $k = \kappa N$ using a differentiable Gumbel-Softmax trick [47] applied to $\hat{w}(\mathbf{r})$; in the forward pass we take the Top- k indices, while in the backward pass we adopt a straight-through surrogate using the soft scores. This yields two complementary subsets, superior subset $\mathbf{r} \odot \mathbf{m}$ and inferior subset $\mathbf{r} \odot (1 - \mathbf{m})$.

$$\mathbf{m} = \text{Gumbel-Softmax}(\hat{w}(\mathbf{r}); \kappa N) \in \mathbb{R}^N, \mathbf{r} \in \{\mathbf{r}, \mathbf{r}^a\},$$

where entries near 1 indicate superior dimensions.

Based on this partition, two masked representations, $\mathbf{r} \odot \mathbf{m}$ and $\mathbf{r} \odot (1 - \mathbf{m})$, are fed to regressors $\hat{h}_1$ and $\hat{h}_2$, respectively. With a regression loss ℓ , two supervision loss terms can be written as follows,

$$\mathcal{L}_{reg}^{sup} = \ell(\hat{h}_1(\mathbf{r} \odot \mathbf{m}), y) + \ell(\hat{h}_1(\mathbf{r}^a \odot \mathbf{m}^a), y),$$

$$\mathcal{L}_{reg}^{inf} = \ell(\hat{h}_2(\mathbf{r} \odot (1 - \mathbf{m})), y) + \ell(\hat{h}_2(\mathbf{r}^a \odot (1 - \mathbf{m}^a)), y).$$

To jointly regulate the contributions of superior and inferior dimensions, we formulate the overall optimization objective for CIR as:

$$\min_{\hat{g}, \hat{h}_1, \hat{h}_2} (\mathcal{L}_{reg}^{sup} + \mathcal{L}_{reg}^{inf} + \tau \mathcal{L}_{FAC}), \quad \min_{\hat{w}} (\mathcal{L}_{reg}^{sup} - \mathcal{L}_{reg}^{inf}),$$

with trade-off parameter $\tau > 0$.

In this min-max formulation, the masker $\hat{w}$ is optimized to find inferior dimensions with less contribution to regression by maximizing $\mathcal{L}_{reg}^{inf}$, whereas the encoder $\hat{g}$ and inferior regressor $\hat{h}_2$ jointly minimize $\mathcal{L}_{reg}^{inf}$ to force the inferior dimensions to carry more causal information beneficial to regression. In parallel, the encoder $\hat{g}$ and superior regressor $\hat{h}_1$ minimize $\mathcal{L}_{reg}^{sup}$, which ensures that the superior dimensions are aligned with the most predictive and dominant causal factors. Through this dual optimization, superior dimensions are encouraged to preserve

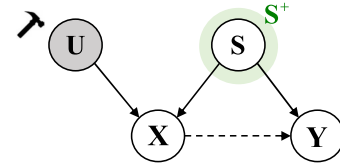


Fig. 5. Structural causal model (SCM) illustrating the role of phase perturbation.

primary regression signals, while inferior dimensions are simultaneously pushed to encode complementary causal information, leading to a causal sufficient and robust representation. Finally, in the inference stage, only the feature extractor $\hat{g}$ and the regressor $\hat{h}_1$ are utilized, which guarantees efficient deployment without additional masker overhead.

IV. CAUSAL ANALYSIS OF PHASE PERTURBATION

A. Necessity of Phase Intervention

The joint amplitude-phase intervention introduced in the AFM module is not arbitrary but motivated by the specific nature of industrial time-series. While in vision tasks the phase spectrum is often assumed to be stable and left unchanged, such an assumption does not hold here. In practice, vibration signals from bearings and related equipment are prone to phase drift, jitter, and timing shifts due to load fluctuations, speed variations, or progressive degradation. These effects undermine the “phase-stable” assumption and make models fragile when encountering unseen conditions [24], [25].

Without considering such variations, amplitude-only augmentation leaves a blind spot: the model never experiences realistic phase-domain shifts during training. Phase perturbation thus complements amplitude intervention by exposing the predictor to operational variations that are both common and unavoidable in industrial settings. This design helps enforce invariance to phase drift and strengthens the robustness of the learned representations under phase-domain shifts.

As illustrated in Fig. 5, conventional approaches often focus on modeling or perturbing the exogenous factors U [23], [48], where U denotes the exogenous (noise) variables in a structural causal model [33], [34]. In contrast, our design explicitly introduces perturbations on S (highlighted as S^+), which directly influences both X and Y . This shift raises the key question of whether such interventions can preserve the label semantics of the downstream task. To answer this, we present two theoretical results: an upper bound based on mutual information from a statistical perspective, and a Lipschitz-type bound from a functional perspective. Together, these provide rigorous support for the controllability of phase perturbation effects.

B. Causal Analysis of Phase Perturbation Effects

This subsection investigates whether the proposed phase perturbation undermines the causal invariance of the degradation mechanism. We first formalize the perturbation in the Fourier domain and define the interventional outcome distributions p_0 (no intervention) and p_δ (phase-perturbed). We then bound the

resulting average causal effect (ACE) from two complementary perspectives: (i) a distributional–information bound that scales with the term related to mutual information, reflecting how distinguishable the two interventions are from the outcome alone; and (ii) a functional Lipschitz bound that scales with the norm $\|\Delta\Phi\|_\infty$ and the spectral energy. Together, these results show that when phase shifts are local, similarity-aware, and bounded—as in our AFM design—the causal effect of phase perturbations remains strictly controlled, thereby preserving causal semantics while improving robustness to realistic phase drift and jitter observed in vibration signals.

Let $\mathbf{x} = \mathcal{F}^{-1}(\mathcal{A} e^{j\mathcal{P}})$ denote a vibration signal, where $\mathcal{F}^{-1}$ represents the inverse Fourier transform. A bounded phase perturbation yields the perturbed signal $\mathbf{x}^a = \mathcal{F}^{-1}(\mathcal{A} e^{j(\mathcal{P} + \Delta\Phi)})$, where $\Delta\Phi = \text{sign}(\Delta\Theta) \Delta\Theta (1 - \lambda_{\mathcal{P}})$. Denote by p_0 and p_δ the conditional output distributions under interventions $\text{do}(\mathbf{x} \leftarrow \mathbf{x})$ and $\text{do}(\mathbf{x}^a \leftarrow \mathbf{x})$, respectively. The ACE of the perturbation is $\text{ACE}_\Phi(\Delta\Phi) = \mathbb{E}_{p_\delta}[Y] - \mathbb{E}_{p_0}[Y]$, where Y is a bounded outcome. We use $\text{TV}(\cdot, \cdot)$ for total variation, $\text{JS}(\cdot, \cdot)$ for Jensen–Shannon divergence, and $I(T; Y)$ for the mutual information between Y and the binary treatment variable T , where $T = 0$ indicates no intervention (p_0) and $T = 1$ indicates intervention with phase perturbation (p_δ).

Assumption 1: Let Y denote the ground-truth RUL label. We assume the label is bounded, i.e., $|Y| \leq C$ for some constant C (which is naturally satisfied for RUL and can be enforced by label normalization).

Theorem 1 (Mutual information bound on ACE): Assume that $|Y| \leq C$, where C denotes a constant number, then the average causal effect admits the mutual-information bound

$$|\text{ACE}_\Phi(\Delta\Phi)| \leq 2C \sqrt{2I(T; Y)}. \quad (6)$$

The proof of Theorem 1 is provided in Appendix A, online available.

Remark 1 (Mutual information perspective): From the viewpoint of information leakage, the mutual information $I(T; Y)$ is inherently small in our setting. By definition, $I(T; Y) = H(Y) - H(Y|T)$. Equivalently, it can be expressed as the expectation of pointwise mutual information (PMI): $I(T; Y) = \sum_{t \in \{0,1\}} \int p(t, y) i(t; y) dy$, $i(t; y) = \log \frac{p(y|t)}{p(y)}$. Here, $i(t; y)$ quantifies the information gain of observing $Y = y$ given treatment $T = t$.

In our design, perturbations are restricted to neighborhoods of samples with similar labels, and their magnitude is adaptively constrained by the label difference. As a result, each $i(t; y)$ contributes only a small value, ensuring that the overall mutual information remains bounded: $I(T; Y) < \varepsilon$, for some small constant $\varepsilon > 0$.

In rotating machinery, phase drift and local jitter naturally arise from load and speed variations, our similarity-aware, shortest-angle phase adjustment mirrors these realistic shifts, keeping $I(T; Y)$ small.

Lemma 1 (Semantic consistency in latent space): According to Proposition 3.1 in [49], if two latent vectors z_1, z_2 satisfy $\sigma(z_1, z_2) \leq \delta$, then their decoded outputs $D(z_1)$ and $D(z_2)$

satisfy

$$|I(D(z_1); Y) - I(D(z_2); Y)| \leq \varepsilon.$$

Remark 2 (Distributional similarity perspective): When phase perturbations correspond to local shifts in the latent space, Lemma 1 ensures that semantic information changes remain bounded. Specifically, the use of bounded and smooth perturbations, together with the semantic-aware similarity metric, effectively limits the mutual information difference between perturbed and original samples. This implies that the perturbed distribution p_δ remains statistically close to the original distribution p_0 , which in turn guarantees a small Jensen–Shannon divergence $\text{JS}(p_\delta, p_0)$. Combined with (6), this leads to the conclusion that the causal effect of phase perturbations is strictly bounded within a controllable small scale.

Assumption 2: There exists a constant $L_f > 0$ such that for all inputs $\mathbf{x}, \mathbf{x}'$, $|f(\mathbf{x}) - f(\mathbf{x}')| \leq L_f \|\mathbf{x} - \mathbf{x}'\|_2$.

Assumption 3: Let $\mathcal{A}(\mathbf{x})$ denote the amplitude spectrum of $\mathbf{x}$. There exists a constant $B_{\mathcal{A}} < \infty$ such that $\|\mathcal{A}(\mathbf{x})\|_2 \leq B_{\mathcal{A}}$ for all $\mathbf{x}$ of interest. Moreover, the phase perturbation satisfies $\|\Delta\Phi(\mathbf{x})\|_\infty \leq \varepsilon_\Phi$ for some $\varepsilon_\Phi < \infty$.

Remark 3: Assumption 2 is a regularity condition ensuring that small input changes induce bounded changes in the prediction. For common neural networks built from linear/convolutional layers and pointwise activations, one may upper bound L_f by the product of layer operator norms and activation Lipschitz constants (e.g., ReLU has constant 1). Assumption 3 is mild for finite-length signals: finite-length (and typically normalized) inputs imply finite spectral energy; the bound $\|\Delta\Phi\|_\infty \leq \varepsilon_\Phi$ can be viewed as enforcing a bounded per-frequency phase update, and can be implemented by clipping or by a bounded step-size rule in the shortest-angle update.

Theorem 2 (Lipschitz–Spectral Bound on ACE): If the prediction function f is L_f -Lipschitz, $\mathcal{A}$ is square-integrable, and C_n denotes the operator norm of the inverse Fourier transform (IFFT), then the average causal effect of a bounded phase perturbation satisfies

$$|\text{ACE}_\Phi(\Delta\Phi)| \leq L_f C_n \|\Delta\Phi\|_\infty \mathbb{E}[\|\mathcal{A}\|_2]. \quad (7)$$

The proof of Theorem 2 can be found in Appendix B, online available.

Remark 4 (Scope of the causal framing): The interventions in this section are defined on the *observed input* $\mathbf{x}$ (i.e., replacing $\mathbf{x}$ by $\mathbf{x}^a$), and the effect is measured on the *predictor output* $Y = f(\mathbf{x})$. Therefore, ACE_Φ quantifies output stability under the designed augmentation, which serves as a robustness/semantic-preservation justification rather than formal causal identification of real-world physical mechanisms from observational data alone.

Remark 5: Equation 7 shows that if $\|\Delta\Phi\|_\infty$ is sufficiently small, then the causal effect of phase perturbations remains small, scaled only by the Lipschitz constant and spectral energy. This aligns with empirical observations that local phase noise induces negligible prediction changes.

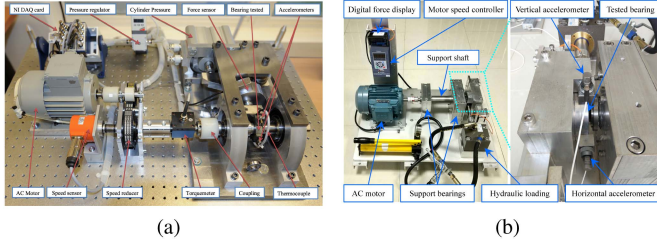


Fig. 6. The PRONOSTIA platform [50] and XJTU-SY accelerated degradation experimental platform [51].

TABLE I
DESCRIPTION OF THE IEEE PHM CHALLENGE 2012 DATASETS

Condition	Speed (rpm)	Load (N)	Bearings
OC1	1,800	4,000	IEEE 1_1 ~ 1_7
OC2	1,650	4,200	IEEE 2_1 ~ 2_7
OC3	1,500	5,000	IEEE 3_1 ~ 3_3

TABLE II
DG TASK SETTING ON THE IEEE PHM CHALLENGE 2012 DATASETS

Task	conditions	Training sets	Unseen target
A1	OC2&3 → OC1	IEEE 2_1, 2_2, 3_1, 3_2	IEEE 1_1, 1_2
A2	OC1&3 → OC2	IEEE 1_1, 1_2, 3_1, 3_2	IEEE 2_1, 2_2
A3	OC1&2 → OC3	IEEE 1_1, 1_2, 2_1, 2_2	IEEE 3_1, 3_2

V. EXPERIMENTAL STUDY

A. Dataset Description and Evaluation Metrics

1) *Case A: IEEE PHM Challenge 2012 Datasets*: The datasets were collected using the PRONOSTIA experimental platform [50], which was specifically designed for accelerated bearing degradation tests. The testbed consists of three modules: a rotation unit, a degradation generation unit, and a data collection unit, as shown in Fig. 6(a). Bearings were operated until failure under different rotational speeds and radial load conditions, simulating diverse operating conditions (OCs), as summarized in Table I. Vibration data were acquired using two orthogonally mounted accelerometers placed in the vertical and horizontal directions. The sensors sampled signals at a frequency of 25.6 kHz, with each measurement spanning 0.1 s, resulting in 2560 data points collected at regular intervals of 10 s.

Following common practice [29], [52], three DG tasks are constructed by training models on bearings from two OCs and evaluating them on unseen bearings from the remaining OC, as detailed in Table II.

2) *Case B: XJTU-SY Run-to-Failure Datasets*: The run-to-failure vibration datasets were collected on the experimental platform shown in Fig. 6(b). Two orthogonally mounted acceleration sensors sampled vibration signals at 25.6 kHz with a duration of 1.28 s, yielding 32,768 data points every minute. To reduce model scale and computational expense, each acquisition segment was further divided into samples of length 2,560.

The dataset includes three OCs, as summarized in Table III. Following the same protocol as in Case A, we construct three multi-source DG tasks, as detailed in Table IV.

TABLE III
DESCRIPTION OF THE XJTU-SY BEARING DATASETS

Condition	Speed (rpm)	Load (kN)	Bearings
OC1	2,100	12	XJTU 1_1 ~ 1_5
OC2	2,250	11	XJTU 2_1 ~ 2_5
OC3	2,400	10	XJTU 3_1 ~ 3_5

TABLE IV
DG TASK SETTING ON THE XJTU-SY BEARING DATASETS

Task	conditions	Training sets	Unseen target
B1	OC2&3 → OC1	XJTU 2_1, 2_2, 3_1, 3_2	XJTU 1_1, 1_2
B2	OC1&3 → OC2	XJTU 1_1, 1_2, 3_1, 3_2	XJTU 2_1, 2_2
B3	OC1&2 → OC3	XJTU 1_1, 1_2, 2_1, 2_2	XJTU 3_1, 3_2

TABLE V
DG TASK SETTING ON THE PHM2010 DATASET

Task	Training sets	Unseen Target
C1	c4, c6	c1
C2	c1, c6	c4
C3	c1, c4	c6

TABLE VI
DEFINITION OF THE CNC MILLING DATASET

Condition	RDOC (mm)	ADOC (mm)	Tools
OC1	4.5	5	8, 9, 102, 105
OC2	4.5	10	7, 10, 101, 103
OC3	8.0	5	2, 4, 6

3) *Case C: PHM2010 Dataset*: The PHM Society 2010 Data Challenge dataset [53] is a widely used benchmark for RUL estimation of high-speed CNC milling cutters. During machining, multi-sensor signals are recorded, including cutting forces, vibrations, and acoustic emission (AE). Each run provides seven synchronized channels: three-axis cutting forces, three-axis vibrations, and the AE-RMS signal, all sampled at 50 kHz.

The dataset contains six cutter runs, denoted as c1–c6. Among them, c1, c4, and c6 include ground-truth wear measurements. We treat each labeled cutter run as an individual domain and adopt a leave-one-domain-out (LODO) protocol, yielding three DG tasks summarized in Table V.

4) *Case D: CNC Milling Dataset*: We further evaluate our method on a recent open CNC milling dataset [54], which includes 14 milling tools and 968 machining cycles collected under real machining conditions. In this work, we only use the vibration signals sampled at 25 kHz.

Domains are defined by machining conditions specified by different combinations of radial depth of cut (RDOC) and axial depth of cut (ADOC), with multiple tools operated under each condition. Following the same multi-source DG protocol as in Cases A & Case B, we construct three DG tasks by training on two source conditions and testing on the remaining unseen condition. To avoid tool leakage, the tools used for training and testing are strictly disjoint. The OC definition and DG task settings are summarized in Tables VI and VII, respectively.

TABLE VII
DG TASK SETTING ON THE CNC MILLING DATASET

Task	Conditions	Training tools	Unseen target
D1	OC2&3 → OC1	2, 4, 6, 7, 10, 101, 103	8, 9, 102, 105
D2	OC1&3 → OC2	2, 4, 6, 8, 9, 102, 105	7, 10, 101, 103
D3	OC1&2 → OC3	7, 8, 9, 10, 101, 102, 103, 105	2, 4, 6

TABLE VIII
NETWORK STRUCTURE OF 1D-UNET BACKBONE

Structure	Output Shape
Input	(B, 2, 2560)
DoubleConv: $2 \times \{\text{Conv1d}(k=3, s=1, p=1) \rightarrow \text{BN} \rightarrow \text{ReLU}\}$	(B, 32, 2560)
Down1: $\text{MaxPool1d}(k=3, s=2, p=1) + \text{DoubleConv}(32 \rightarrow 64)$	(B, 64, 1280)
Down2: $\text{MaxPool1d}(k=3, s=2, p=1) + \text{DoubleConv}(64 \rightarrow 128)$	(B, 128, 640)
Down3: $\text{MaxPool1d}(k=3, s=2, p=1) + \text{DoubleConv}(128 \rightarrow 256)$	(B, 256, 320)
Bottleneck: $\text{Conv1d}(1 \times 1, 256 \rightarrow 64) + \text{AdaptiveAvgPool1d}(1)$	(B, 64)
Up1: $\text{Upsample}(\times 2) + \text{Conv1d}(1 \times 1, 256 \rightarrow 128) + \text{Concat} + \text{DoubleConv}(256 \rightarrow 128)$	(B, 128, 640)
Up2: $\text{Upsample}(\times 2) + \text{Conv1d}(1 \times 1, 128 \rightarrow 64) + \text{Concat} + \text{DoubleConv}(128 \rightarrow 64)$	(B, 64, 1280)
Up3: $\text{Upsample}(\times 2) + \text{Conv1d}(1 \times 1, 64 \rightarrow 32) + \text{Concat} + \text{DoubleConv}(64 \rightarrow 32)$	(B, 32, 2560)
OutConv: $\text{Conv1d}(1 \times 1, 32 \rightarrow 2)$	(B, 2, 2560)

5) *Evaluation Metrics*: To comprehensively evaluate the prediction performance of the proposed method, we adopt three widely used metrics: mean absolute error (MAE), root mean squared error (RMSE), and scoring function.

$$MAE = \frac{1}{n} \sum_{i=1}^n |y_i - \hat{y}_i|, \quad RMSE = \sqrt{\frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2}$$

$$Score = \frac{1}{n} \sum_{i=1}^n A_i,$$

where

$$A_i = \begin{cases} e^{-\ln(0.5) \cdot (Er_i/5)}, & \text{if } Er_i \leq 0 \\ e^{+\ln(0.5) \cdot (Er_i/20)}, & \text{if } Er_i > 0 \end{cases}$$

where Er_i is the percentage error:

$$Er_i = \frac{y_i - \hat{y}_i}{y_i} \times 100,$$

where y_i and $\hat{y}_i$ are the ground-truth label and predicted values, respectively. Smaller deviations from the ground truth lead to lower MAE and RMSE, and consequently higher scores. These three metrics offer a comprehensive evaluation, capturing not only predictive accuracy but also the temporal reliability of prognostic estimates.

B. Implementation Details

With the datasets and evaluation metrics established, we now introduce the overall architecture of our proposed framework.

To construct the domain-invariant order-preserving guidance embedding, we adopt a 1D-UNet encoder-decoder with skip connections. The encoder and decoder comprise stacks of 1D convolutional blocks interleaved with pooling and up-sampling operations, as detailed in Table VIII. To more effectively model the two-channel vibration signals, we incorporate multi-kernel convolutions and residual connections, which expand the receptive field and enhance optimization stability. The domain discriminator $D(\cdot; \phi)$ is implemented as a randomly initialized two-layer MLP.

TABLE IX
HYPERPARAMETER SETTING

Parameter	Value	Parameter	Value
learning rate	5e-5	$\gamma_{\mathcal{A}}$	0.5
training epochs	100	$\gamma_{\mathcal{P}}$	0.9
optimizer	Adam	η	0.8
batch size	256	κ	0.8
α_{adv}	1	τ	2
α_{RNC}	1		

For the CIR module, the feature extractor is a 1D-ResNet backbone with a hidden feature dimension of $N = 512$, while two regressors are implemented as a linear layer with sigmoid activation, and the masker $\hat{w}$ is realized as a two-layer MLP with dropout, batch normalization, and ReLU, followed by iterative Gumbel-softmax sampling to generate adaptive feature masks.

All experiments were implemented in Python 3.10 using the PyTorch 2.1 framework, and executed on a server equipped with two NVIDIA A40 GPUs (40 GB memory each) and an AMD EPYC 7H12 64-core processor running at 3.10 GHz. To reduce the effect of randomness, prediction results are averaged over 10 independent runs. During training, the Adam optimizer with a decayed learning rate schedule is adopted to ensure stable convergence. Hyperparameters are tuned via grid search cross-validation, and the final configuration is summarized in Table IX. Note that for the PHM 2012 and XJTU-SY bearing datasets, the model input consists of two-channel vibration signals.

C. Main Result

To verify the effectiveness of the proposed method, we compare it against several representative and widely used RUL prediction methods. The comparative methods cover three categories: deep learning-based RUL methods, general DG methods, and DG-based RUL prediction methods.

- *Deep learning-based RUL methods*: MSCNN [55], DCNN [56], and CLSTM [57], which rely on convolutional or recurrent architectures and are trained on multi-source domains before directly testing on unseen targets.
- *general DG methods*: Mixup [12], RSC [58], and L2A-OT [59], which are representative methods on data manipulation and training strategies. Though not designed specifically for RUL prediction, they are widely adopted in DG tasks.
- *DG-based RUL methods*: AOA [52], STITA-DANN [44], and MODFE [29], which integrate DG with RUL prediction through data manipulation and domain-invariant representation learning.

These counterparts together provide a comprehensive basis for evaluating both the predictive accuracy and DG ability of the proposed method.

Table X summarizes the performance of the proposed method against a range of state-of-the-art counterparts on two publicly available bearing datasets. All numbers are reported as the overall average across the 12 DG transfer tasks from Case A&B, with mean $\pm$ std shown when available. Several key observations can be made. First, conventional DL-based RUL

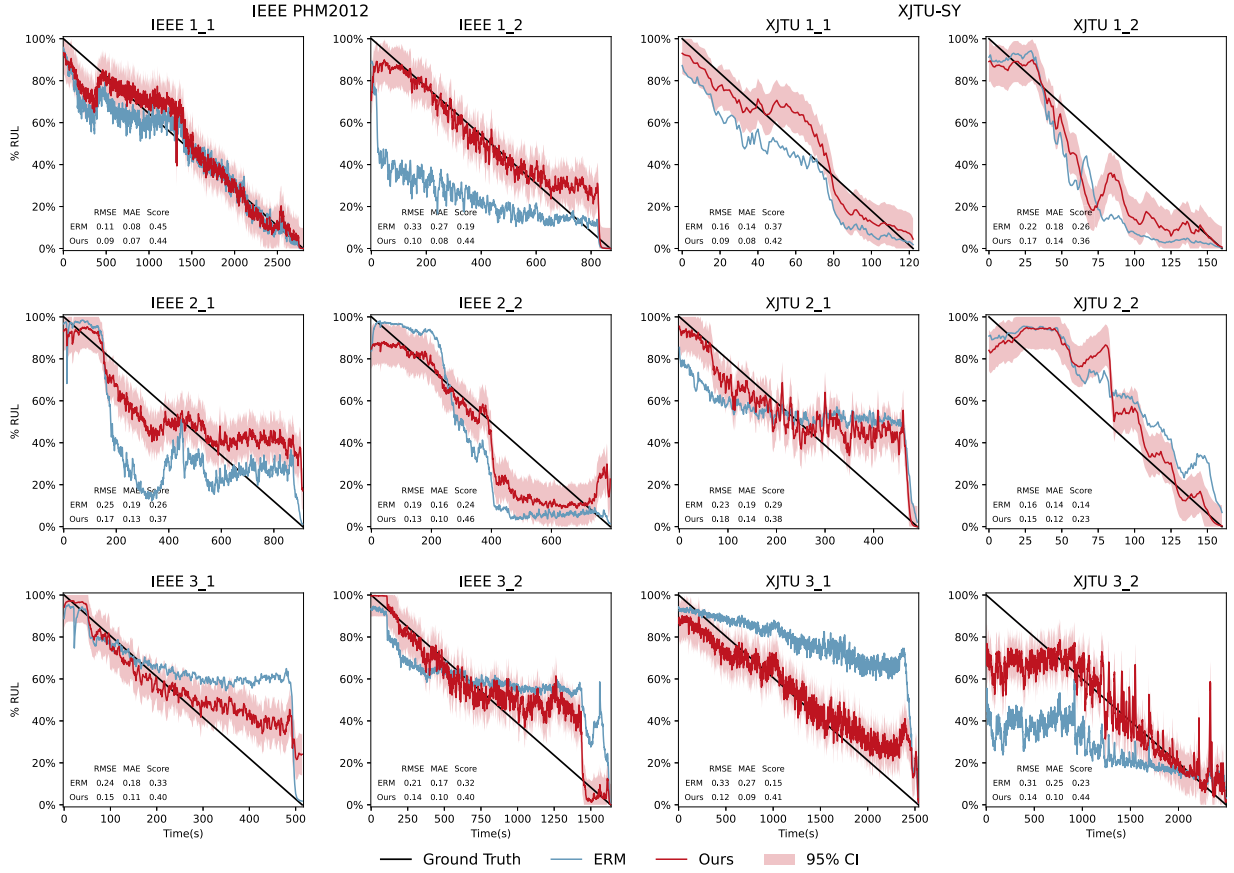


Fig. 7. RUL prediction results of ERM baseline and AFM-CIR for all the DG tasks of Case A & Case B.

TABLE X
COMPARISON RESULT ON CASE A & CASE B, AVERAGED OVER 12 DG TASKS

Method	RMSE	MAE	Score
MSCNN [55]	0.3178	0.2595	0.2209
DCNN [56]	0.2911	0.2394	0.2058
CLSTM [57]	0.2534	0.2013	0.2835
Mixup [12]	0.2496	0.2025	0.2452
RSC [58]	0.2125	0.1695	0.2739
L2A-OT [59]	0.1867	0.1441	0.2975
AOA [52]	0.1590±0.0150	0.1296±0.0124	0.3372±0.0201
STITA-DANN [44]	0.1525±0.0157	0.1233±0.0129	0.3648±0.0315
MODFE [29]	0.1486±0.0164	0.1198±0.0131	0.3747±0.0365
AFM-CIR	0.1358±0.0157	0.1043±0.0136	0.3959±0.0184

models trained under empirical risk minimization (ERM) exhibit limited generalization ability, confirming that directly training on source domains without explicit DG mechanisms struggles to handle unseen target conditions. Second, general DG approaches yield noticeable improvements over the DL-based models, highlighting the importance of DG strategies in mitigating distribution shifts. Third, DG-based RUL prediction methods further enhance prediction accuracy, surpassing all general DG methods. Finally, the proposed AFM-CIR achieves the best results across all metrics and datasets, demonstrating the advantage of combining adaptive Fourier augmentation

with causality-inspired regression. These improvements establish AFM-CIR as an effective solution for cross-domain RUL prediction.

Fig. 7 presents the RUL prediction curves of AFM-CIR and ERM baseline on the IEEE and XJTU-SY bearing datasets. In the figure, the black curve denotes the ground-truth label, while the blue curve represents predictions from the ERM baseline, i.e., the same backbone network trained on multi-source domains under the ERM objective to generalize to unseen target domains, but without any of the proposed AFM-CIR components. The red curve shows the predictions of the proposed AFM-CIR and the light red shaded area indicates the 95% confidence interval (CI), calculated based on the standard deviation (std) of 10 prediction results. It is worth noting that the prediction curves in Fig. 7 were moving-averaged only for visualization purposes to smooth excessive fluctuations, whereas the raw predictions are used for the actual performance metric calculations.

The averaged results together with the std over 10 independent runs on the bearing benchmarks are summarized in Table XI. The results clearly show that the proposed method achieves more accurate and stable RUL predictions than the ERM baseline across most degradation tasks. In particular, the predicted trajectories of AFM-CIR align more closely with the ground-truth degradation trends, especially during the middle and late stages, where accurate forecasting is crucial for timely maintenance.

Performance improvements, however, are not uniform across all tasks. Substantial improvements are observed for IEEE 1_2,

TABLE XI
TASK-WISE MEAN $\pm$ STD OF PREDICTED RUL RESULTS FOR ERM BASELINE AND AFM-CIR ON CASE A & CASE B

Task	ERM Baseline			AFM-CIR		
	RMSE	MAE	Score	RMSE	MAE	Score
IEEE 1_1	0.1237 $\pm$ 0.0319	0.0996 $\pm$ 0.0310	0.4374 $\pm$ 0.0638	0.0871 $\pm$ 0.0147	0.0664 $\pm$ 0.0128	0.4427 $\pm$ 0.0275
IEEE 1_2	0.2911 $\pm$ 0.0478	0.2429 $\pm$ 0.0446	0.2171 $\pm$ 0.0529	0.0993 $\pm$ 0.0196	0.0774 $\pm$ 0.0184	0.4446 $\pm$ 0.0243
IEEE 2_1	0.2829 $\pm$ 0.0598	0.2229 $\pm$ 0.0487	0.2328 $\pm$ 0.0481	0.1705 $\pm$ 0.0227	0.1343 $\pm$ 0.0195	0.3703 $\pm$ 0.0212
IEEE 2_2	0.2136 $\pm$ 0.0446	0.1738 $\pm$ 0.0354	0.2194 $\pm$ 0.0361	0.1325 $\pm$ 0.0169	0.1010 $\pm$ 0.0143	0.4578 $\pm$ 0.0178
IEEE 3_1	0.2501 $\pm$ 0.0225	0.1825 $\pm$ 0.0169	0.3128 $\pm$ 0.0340	0.1516 $\pm$ 0.0104	0.1141 $\pm$ 0.0087	0.4049 $\pm$ 0.0151
IEEE 3_2	0.2301 $\pm$ 0.0175	0.1862 $\pm$ 0.0130	0.2940 $\pm$ 0.0203	0.1374 $\pm$ 0.0081	0.0982 $\pm$ 0.0070	0.4031 $\pm$ 0.0096
XJTU 1_1	0.1530 $\pm$ 0.0210	0.1264 $\pm$ 0.0194	0.3863 $\pm$ 0.0421	0.0916 $\pm$ 0.0134	0.0775 $\pm$ 0.0108	0.4201 $\pm$ 0.0237
XJTU 1_2	0.2332 $\pm$ 0.0160	0.1905 $\pm$ 0.0129	0.2495 $\pm$ 0.0186	0.1722 $\pm$ 0.0097	0.1352 $\pm$ 0.0086	0.3564 $\pm$ 0.0128
XJTU 2_1	0.2809 $\pm$ 0.0397	0.2272 $\pm$ 0.0282	0.2555 $\pm$ 0.0279	0.1793 $\pm$ 0.0186	0.1359 $\pm$ 0.0152	0.3786 $\pm$ 0.0171
XJTU 2_2	0.1891 $\pm$ 0.0417	0.1651 $\pm$ 0.0371	0.1472 $\pm$ 0.0307	0.1467 $\pm$ 0.0165	0.1160 $\pm$ 0.0139	0.2261 $\pm$ 0.0131
XJTU 3_1	0.2383 $\pm$ 0.0408	0.1916 $\pm$ 0.0396	0.2083 $\pm$ 0.0490	0.1190 $\pm$ 0.0178	0.0944 $\pm$ 0.0163	0.4074 $\pm$ 0.0189
XJTU 3_2	0.3365 $\pm$ 0.0482	0.2746 $\pm$ 0.0449	0.1991 $\pm$ 0.0470	0.1425 $\pm$ 0.0202	0.1012 $\pm$ 0.0176	0.4388 $\pm$ 0.0197
Average	0.2352 $\pm$ 0.0360	0.1903 $\pm$ 0.0310	0.2633 $\pm$ 0.0392	0.1358 $\pm$ 0.0157	0.1043 $\pm$ 0.0136	0.3959 $\pm$ 0.0184

TABLE XII
TASK-WISE MEAN $\pm$ STD OF PREDICTED RUL RESULTS FOR THE ERM BASELINE AND AFM-CIR ON CASE C

Task	ERM Baseline		AFM-CIR	
	RMSE	MAE	RMSE	MAE
C1	0.0901 $\pm$ 0.0042	0.1087 $\pm$ 0.0051	0.0600 $\pm$ 0.0033	0.0711 $\pm$ 0.0038
C2	0.0343 $\pm$ 0.0026	0.0436 $\pm$ 0.0030	0.0289 $\pm$ 0.0022	0.0417 $\pm$ 0.0029
C3	0.0411 $\pm$ 0.0029	0.0530 $\pm$ 0.0035	0.0270 $\pm$ 0.0021	0.0373 $\pm$ 0.0027
Average	0.0552 $\pm$ 0.0032	0.0684 $\pm$ 0.0039	0.0386 $\pm$ 0.0025	0.0500 $\pm$ 0.0031

3_1, 3_2, and XJTU 3_1, 3_2, where both RMSE and MAE are substantially reduced and the scoring function improves markedly. Moderate but consistent gains appear in IEEE 2_1, XJTU 1_2, 2_1, and 2_2 while the performance on IEEE 1_1 remains close to the baseline due to already satisfactory predictions.

To further validate the generalization of AFM-CIR beyond bearing degradation scenarios, we additionally evaluate it on two milling datasets (Case C & D). Compared with the bearing benchmarks, these milling signals are collected under substantially different sensing configurations, including different sampling rates and higher-dimensional multivariate measurements. These additional experiments therefore probe the robustness of AFM-CIR to variations in sampling frequency and sensor dimensionality while keeping the overall goal of cross-domain RUL prediction unchanged. Since these benchmarks are less standardized for direct SOTA comparison under our DG protocol, we report results for the ERM baseline and AFM-CIR under the same training recipe as in Case A & B. Unless otherwise specified, all hyperparameters are inherited from the validated configuration in Table IX without additional per-dataset tuning.

On PHM2010 (Case C), Table XII shows that AFM-CIR achieves lower RMSE/MAE than ERM baseline on all three DG tasks and on average. Fig. 8 complements this finding: compared with ERM (blue), AFM-CIR (red) produces predictions that align more closely with the ideal diagonal, indicating reduced bias and improved accuracy under domain shift.

On the CNC dataset (Case D), Table XIII reports consistent gains of AFM-CIR over ERM across all tasks, together with reduced variability. Fig. 9 further visualizes the prediction

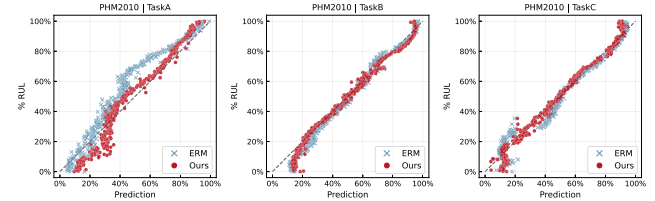


Fig. 8. RUL prediction scatter plots on Case C comparing the ERM baseline and AFM-CIR. Blue points denote ERM predictions and red points denote AFM-CIR predictions. The dashed diagonal indicates perfect prediction.

TABLE XIII
TASK-WISE MEAN $\pm$ STD OF PREDICTED RUL RESULTS FOR THE ERM BASELINE AND AFM-CIR ON CASE D

Task	ERM Baseline		AFM-CIR	
	RMSE	MAE	RMSE	MAE
D1	0.1603 $\pm$ 0.0128	0.1952 $\pm$ 0.0156	0.1452 $\pm$ 0.0116	0.1788 $\pm$ 0.0143
D2	0.2272 $\pm$ 0.0182	0.2662 $\pm$ 0.0213	0.1928 $\pm$ 0.0154	0.2301 $\pm$ 0.0184
D3	0.1279 $\pm$ 0.0102	0.1607 $\pm$ 0.0129	0.1070 $\pm$ 0.0086	0.1385 $\pm$ 0.0111
Average	0.1718 $\pm$ 0.0137	0.2074 $\pm$ 0.0166	0.1483 $\pm$ 0.0119	0.1825 $\pm$ 0.0146

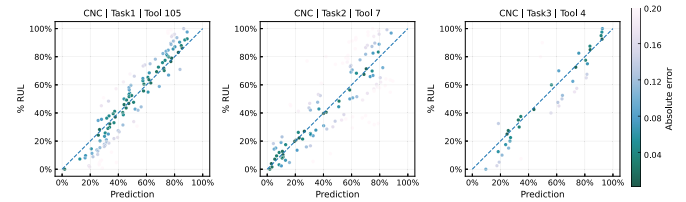


Fig. 9. Absolute-error colored scatter plots of AFM-CIR on the Case D. The dashed diagonal denotes perfect prediction, and the color indicates the absolute error between prediction and ground truth.

behavior, where most samples lie close to the diagonal with relatively small absolute errors, suggesting reliable cross-domain RUL estimation under tool- and condition-dependent distribution shifts.

Overall, AFM-CIR delivers consistent improvements over ERM on both bearing and milling benchmarks, supporting its robustness to domain shifts under different sampling rates and multivariate sensing configurations.

TABLE XIV
ABLATION STUDY OF DIFFERENT MODULE

Method	AFM.	CFac.	AdvM.	RMSE	MAE	Score
ERM Baseline	—	—	—	0.2352	0.1903	0.2633
Variant 1	✓	—	—	0.1836	0.1449	0.3018
Variant 2	—	✓	—	0.1806	0.1467	0.3133
Variant 3	✓	✓	—	0.1737	0.1373	0.3266
Variant 4	✓	—	✓	0.1757	0.1357	0.3389
AFM-CIR	✓	✓	✓	0.1358	0.1043	0.3959

TABLE XV
ABLATION COMPARING MIXING STRATEGIES AND WEIGHTING SCHEME

Strategy	Weighting	RMSE	MAE	Score
Linear Mixup	random	0.2108	0.1689	0.2878
	adaptive	0.1969	0.1560	0.3008
Amplitude Mixing	random	0.1806	0.1467	0.3133
	adaptive	0.1697	0.1362	0.3260
AFM	adaptive	0.1358	0.1043	0.3959

D. Ablation Study

1) *Module Contribution*: Table XIV evaluates the impact of AFM, the factorization loss (CFac), and the adversarial masker (AdvM). The ERM baseline yields the weakest results, underscoring limited cross-domain generalization. Introducing AFM alone (Variant 1) improves over the baseline, indicating that richer, causality-aware fourier augmentation benefits training. Using amplitude-mixing within the CIR framework without AFM (Variant 2) further improves upon the baseline but remains inferior to configurations that incorporate our AFM module. When either causal component is removed, AdvM in Variant 3 or CFac in Variant 4, performance degrades relative to the combined settings, evidencing that enforcing the Common Cause and ICM principles via CFac and ensuring causal sufficiency via AdvM are both necessary. The full model achieves the best prediction performance, demonstrating that AFM coupled with CIR is crucial for robust cross-domain RUL prediction.

2) *Mixup Strategy and Weighting Scheme*: Table XV reveals two consistent trends. First, frequency-domain amplitude mixing outperforms time-domain linear mixup under the same weighting scheme. This aligns with causal considerations: amplitude mixing perturbs magnitudes without synthesizing phase-inconsistent waveforms, whereas time-domain mixing can create non-physical interference that erodes temporal causality. Second, similarity-based adaptive weighting uniformly improves over random weighting within each strategy, because it down-weights dissimilar pairs and avoids injecting destructive information that distorts temporal structure.

Nevertheless, pure amplitude mixing offers limited diversity, the augmented signals remain close to their origin. Our AFM addresses this by coupling similarity-guided amplitude mixing with controlled phase perturbations, thereby expanding the effective support of training variations while preserving causal semantics. As a result, AFM achieves the strongest overall performance across metrics and datasets.

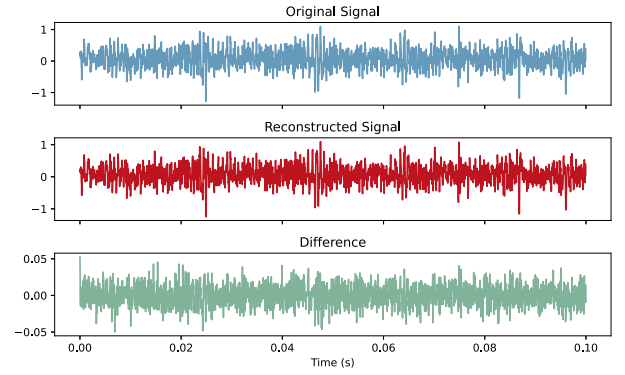


Fig. 10. Reconstruction performance of domain-invariant order-preserving guidance embedding space on a sample signal from the unseen IEEE 3_2 in the PHM 2012 dataset.

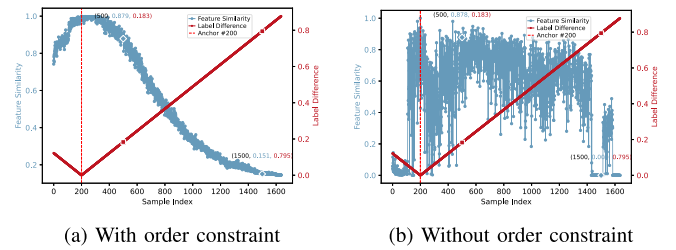


Fig. 11. Feature similarity versus label difference on the PHM 2012 dataset (IEEE 3_2 as unseen target domain), shown with and without the order-preserving constraint $\mathcal{L}_{RNC}$.

E. Empirical Analysis

1) *Reconstruction and Order Preservation*: As illustrated in Fig. 10, the 1D-UNet autoencoder reconstructs the input signal with high fidelity, showing only negligible errors relative to the original waveform. This confirms the model's strong signal reconstruction capability. Beyond reconstruction, we further imposes an order-preserving constraint on the latent space to ensure that embedding similarities are aligned with relative label differences. As shown in Fig. 11, the constraint enforces a smooth decrease of similarity as label differences increase, whereas removing it leads to irregular and inconsistent patterns. These results demonstrate that the order constraint enables the learned similarity to serve as a reliable foundation for the AFM module.

2) *Spectrum Analysis*: To further investigate the effect of different mixup strategies, we conduct spectrum analysis on three representative samples from the IEEE 3_2 bearing, which contains 1,637 vibration signals in total: the anchor signal at cycle 200, a closely related segment at cycle 500, and a distant segment at cycle 1,500. Fig. 12 shows the original time domain signals and their corresponding amplitude spectra. The specific label differences and similarity values between these segments are provided in Fig. 11. Note that for this analysis, we intentionally use within-domain samples instead of cross-domain ones, since cross-domain spectra often exhibit substantial distributional shifts that would obscure the effect of degradation progression. This controlled setting eliminates

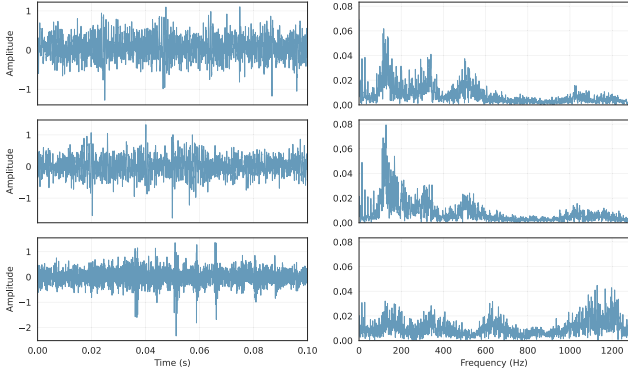


Fig. 12. Time domain horizontal signal and corresponding amplitude spectra of three segments from IEEE 3_2. (a) anchor at cycle 200; (b) neighbor sample at cycle 500; (c) reference sample at cycle 1500.

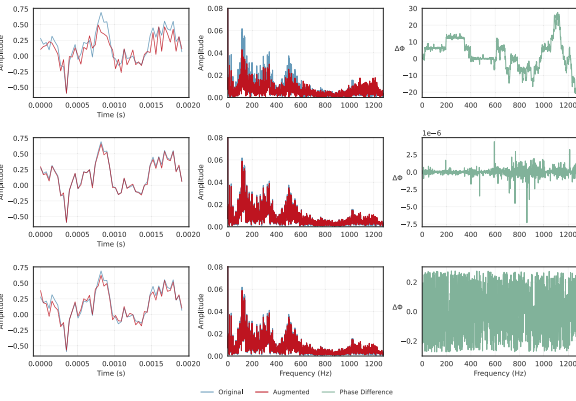


Fig. 13. Comparison of three different augmentation strategies: (top) time domain linear mixup, (middle) frequency domain amplitude-only mixing, and (bottom) the proposed AFM module.

domain-induced variability and enables us to focus exclusively on the spectral differences driven by degradation.

Fig. 13 compares three augmentation strategies: time domain linear mixup, frequency domain amplitude-only mixing, and the proposed AFM module. Each row corresponds to one strategy, while three columns respectively show the anchor and mixed signals in the time domain, the amplitude spectra before and after augmentation, and the phase differences. For a fair comparison, the mixing coefficient (λ_A for temporal and amplitude mixing, λ_P for phase perturbation) are determined using the similarity-based weighting scheme.

As shown in Fig. 13, all three methods introduce noticeable variations in both time and frequency domains. However, temporal linear mixup distorts the causal structure and produces inconsistent temporal trends: the original signal increases while the mixed signal decreases in the same region. By contrast, amplitude-only mixing and AFM preserve the temporal progression of the anchor, with differences mainly reflected in oscillation magnitudes. In the frequency domain, linear mixup causes larger distortions in the low- and mid-frequency ranges and introduces irregular phase deviations with abrupt jumps. Amplitude-only mixing, however, produces almost no phase perturbation, limiting robustness against phase-domain shifts. The proposed AFM achieves a balance: it maintains causal

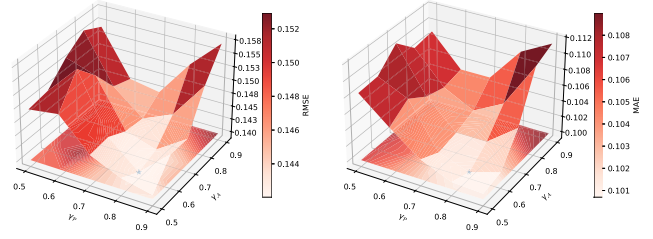
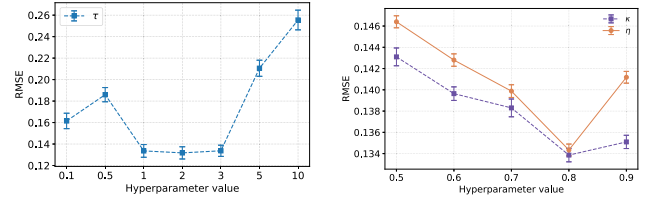


Fig. 14. Sensitivity of AFM-CIR to the hyperparameters γ_A and γ_P . All results are conducted on the PHM 2012 dataset with IEEE 3_2 as the unseen target domain.



(a) Coefficient of factorization loss τ (b) Superior dimension ratio κ and neighbor-pool threshold η

Fig. 15. Sensitivity of AFM-CIR to the hyperparameters on Case A (IEEE 3_2 as unseen target domain).

semantics while introducing small, controlled phase perturbations that preserve smoothness and enhance robustness.

Overall, the spectrum analysis demonstrates that AFM introduces meaningful yet controlled perturbations. It preserves temporal consistency, regulates amplitude variations according to similarity, and injects smooth, causality-preserving phase perturbations. As a result, the augmented signals are both realistic and more robust against distribution shifts in the phase domain.

F. Hyperparameter Tuning

To assess the sensitivity of the proposed AFM-CIR framework to hyperparameter choices, we evaluate five critical hyperparameters: (i–ii) γ_A and γ_P , which specify the lower bounds of adaptive mixup coefficients for amplitude mixing and phase perturbation in the AFM module, respectively; (iii) τ , the coefficient of the factorization loss that enforces feature causal independence and consistency in the CIR module; (iv) κ , the proportion of superior latent dimensions retained; and (v) η , the neighbor-pool similarity threshold for selecting candidates in phase perturbation;

Fig. 14 shows the effect of γ_A and γ_P . We observe that stronger amplitude perturbations (i.e., smaller γ_A) significantly enhance RUL prediction by increasing sample diversity while maintaining structural consistency. Hence, γ_A is set to 0.5. In contrast, phase perturbation requires stricter control: larger values of γ_P (i.e., weaker perturbations) consistently yield better results, since phase governs causal and semantic information. Accordingly, γ_P is set to 0.9 to introduce minimal yet effective phase variations.

As shown in Fig. 15, both τ and κ exhibit a U-shaped relationship with prediction error: overly small values under-regularize the latent space, while excessively large values impose

unnecessary constraints and hinder generalization. The best trade-off is achieved at $\tau = 2$ and $\kappa = 0.8$, which yield the lowest RMSE. We further evaluate the neighbor-pool threshold η in Fig. 15(b). A large η produces an overly small pool and reduces augmentation diversity, while a small η enlarges the pool but may admit semantically mismatched neighbors, introducing detrimental phase noise. Overall, an intermediate choice works best, and $\eta = 0.8$ achieves the lowest error in our experiments.

In summary, AFM-CIR is influenced by both augmentation-control and causal-learning hyperparameters. Proper tuning of γ_A , γ_P , η , τ , and κ is important for balancing augmentation diversity, semantic consistency, and causal regularization, thereby achieving strong cross-domain generalization.

VI. CONCLUSION

This paper introduced AFM-CIR, a framework that jointly injects causal constraints at both the data augmentation and learning objective levels. On the augmentation side, a domain-invariant, order-preserving guidance embedding enables similarity-adaptive amplitude mixing and bounded phase intervention, thereby simulating realistic phase shifts with minimal semantic distortion. On the learning side, correlation factorization enforces invariance and inter-dimensional independence, while adversarial masking improves robustness by ensuring causal sufficiency across weakly informative dimensions. Experimental results on four industrial benchmarks confirm that AFM-CIR consistently outperforms competitive baselines, while ablation studies reveal that random pairing, amplitude-only augmentation, or regression without causal constraints are insufficient. From a theoretical perspective, we established the necessity and controllability of phase interventions by bounding their effects on mutual information and prediction consistency through Lipschitz-spectral norms.

Overall, the integration of semantic similarity-aware frequency augmentation with causality-inspired regression offers a principled and controllable path toward reliable out-of-domain generalization in time-series tasks. Importantly, the idea of AFM is not confined to causal frameworks: the core principle of semantic preservation coupled with structural regularization can naturally extend to broader mixup strategies, thereby opening up new possibilities for representation learning and robust generalization even under non-causal settings.

Looking ahead, we see two promising directions: extending AFM-CIR beyond regression to broader time-series tasks, and further advancing the theoretical analysis of phase interventions and their interplay with domain-invariant mechanisms. We expect the proposed combination of semantics-aware augmentation and causal regression learning to provide a principled path toward reliable, out-of-domain generalization across diverse time-series applications.

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